

A Voyage to the Kernel



Part 9

Segment – 2.2, Day 8

In the last column, we had discussed some basic algorithms and methodologies. Now we will generalise the scheme of an algorithm. It will have the following properties:

- Arranged in an ordered sequence that you can number (as steps)
- Unambiguous and well-defined
- Halts in finite time (i.e., the algorithm terminates!)

And we will generally stumble upon the following algorithmic operations:

- Sequential operations—where instruction sets are executed in order
- Conditional operations—that ask a true/false question and then select the next instruction based on the answer
- Iterative operations (loops)—that repeat the execution of a set of instructions

It should be emphasised that, as we discussed before, not every problem has a ‘good’ algorithmic solution. There are:

- Halting problems (unsolvable problems)—for which no algorithm exists to solve the problem.
- Travelling salesman problems (intractable problems)—algorithms that take too long to solve the problem.
- Problems with no known algorithmic solution.

Pseudo code structure

The algorithm may contain the following elements: IF-THEN-ELSE

Binary choice on a given Boolean condition

is indicated by: IF, THEN, ELSE, and ENDIF. The general representation is:

IF condition THEN

sequence 1

ELSE

sequence 2

ENDIF

Please note that the ELSE keyword and “sequence 2” are optional.

WHILE: The loop is executed only if the condition is true. The ‘WHILE construct’ is used to specify a loop with a test at the top. WHILE and ENDWHILE are to identify the beginning and ending of the loop. The general form for WHILE is:

WHILE condition

sequence

ENDWHILE

CASE: In order to handle mutually exclusive conditions, we can go for CASE. CASE, OF, OTHERS, and ENDCASE are the keywords commonly used for this. The alternatives available will be represented with the help of ‘conditions’.

CASE expression OF